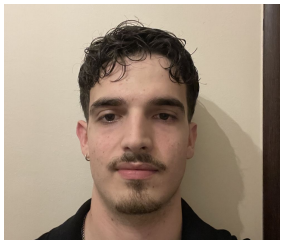
BigO No

Real-time Greek NLP Aggregator Pipeline for Political Bias

Team Members (CEID - University Of Patras):

✦ Elmos Stauridis ✦ Nikos Derekenaris ✦ Vaggelis Mitrogiannis ✦ Kostas Loukanaris



Our Mentors:

✦ Yash Gupta (**NVIDIA**) ✦ Alexandros Liapatis (**KIEFER**)




OpenACC More Science, Less Programming

OPEN HACKATHONS

Hellenic Insight

- **Project Overview**

Real-Time Greek News Aggregator Pipeline for detecting ideological framing and contextualizing extreme political views

- **Problem**

To raise awareness by revealing an article's hidden insight, especially useful for Greece where media is deeply influenced by political interests

- **Tech Stack & AI Model**

Kafka, Spark, MongoDB for data processing/communication and Qwen 3.6-35B (Fine-Tuned) for inference

- **Focus during hackathon**

High accuracy prediction on where the article is politically standing using a contemporary tech stack for future probability of deployment

Evolution and Strategy

- **Goal coming into the hackathon**

To build a basic proof-of-concept tool that could read a Greek article and guess its political leaning, a project one of its kind in Greece. But also learn, connect with new people and have a bit of fun out of our comfort zone

- **Initial strategy**

Came in having bits of the pipeline ready, with our focus on efficient assembling and robust results, both for our model and app

- **How did the initial strategy change?**

Despite facing issues the strategy didn't change , each team member was focused on his assigned part and with the help of our mentors and teamwork we managed to achieve our goals

Progress

Day 0: The Starting Point

- **Architecture Finalized:** Established a robust baseline bridging the Kafka as a message queue with PySpark data ingestion with a Node.js API and React dashboard.
- **Active Streaming:** Automated scraping pipeline successfully live for two primary Greek news sources.
- **Model Fine-Tuned:** Qwen 2.5 14B optimized with custom LoRA on a specific political bias dataset.
- **Target Locked:** Preparing to integrate the vLLM inference layer and scale to 100+ concurrent real-time requests.

Progress

Day 1: Benchmarking & Integration

- **Model Evaluation:** Benchmarked Qwen, Kri-Kri, and Gemma on GreekMMLU. Qwen identified as top-tier with **82.9% 3-shot accuracy**.
- **Optimization:** Deployed a high-efficiency fine-tuning pipeline using **Unsloth Docker** images for rapid iteration.
- **Real-time Analysis:** Implemented the **"Scrape4AI"** module, enabling immediate "Link-to-Insight" functionality for Greek news URLs.
- **Infrastructure:** Exposed vLLM server endpoints and updated core libraries for full **NVIDIA B200 (Blackwell)** compatibility.

Progress

Day 2: Step by Step Improvements

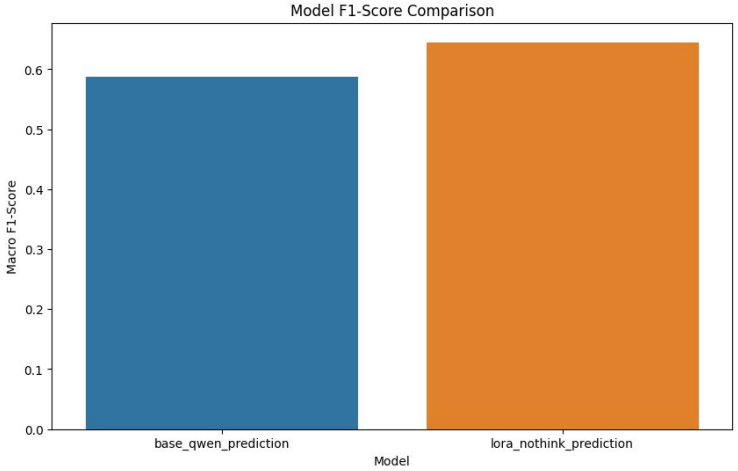
- **Advanced Fine-Tuning:** Optimized **Qwen 3.6 35B** using LoRA on a curated dataset of 2,000 Greek political articles.
- **Robust Pipeline:** Successfully integrated the full data flow: **Static JSON → Kafka → PySpark → vLLM → MongoDB**.
- **Feature Engineering:** Refactored the "Paste Link" feature into a **persistent aiohttp server**, significantly reducing latency via browser warming.
- **Scalability:** Increased active scraping sources to 5, ensuring comprehensive coverage of the Greek media landscape.

Progress

Day 3: End-to-End Success

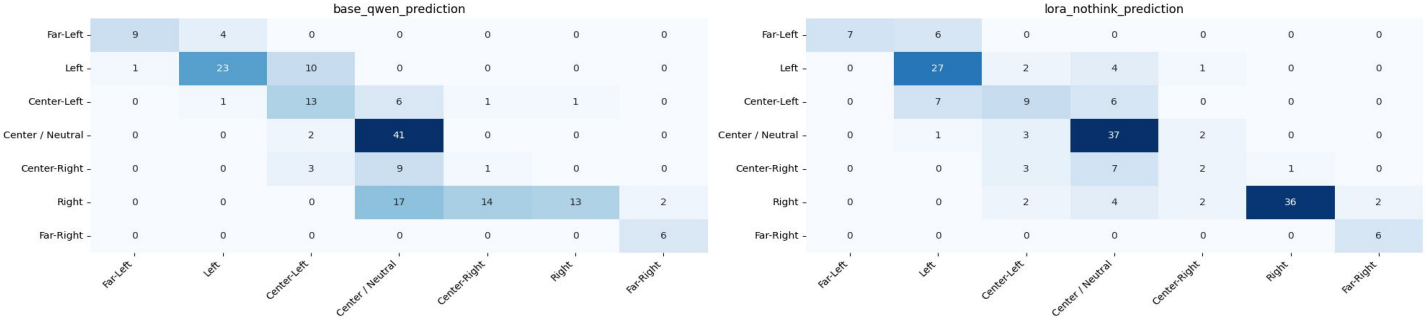
- **Live Pipeline Bridge:** Successfully synchronized the Live Scraper with the main processing pipeline for true real-time analysis.
- **Metric Validation:** Extracted core KPIs proving the model's high precision in political ideological grading and sentiment detection.
- **Global Readiness:** Enhanced the dashboard with **English language support** and granular metadata filters for deep-dive exploration.
- **Production Deployment:** Application fully tested, deployed, and live—**the final demo is ready for presentation.**

Results and Final Profile



- Fine-tuned version improves on classification accurarcy (**~10% F1 boost)**
- Confusion matrix shows base model retreated to neutrality when faced with complexity *(17 missclassified right articles as Center/Neutral)*
- Our custom model successfully diagonalized the confusion matrix, an **~275% improvement** in that specific category

Note the dramatic improvement in 'Right' and 'Center-Right' classification accuracy



Problems We Encountered

- Slow internet connection didn’t allow us to quickly test models on Day 1
- Difficulty in finding a dataset to satisfiably fine-tune our model
- B200 GPUs contained contemporary library versions, which made benchmarking our models messy on GreekMMLU
- Scraping on single articles was challenging, as certain Greek news websites have bots to deployed which restricted data extraction
- Frontend/Backend Bugs

Wishlist

- The whole coordination and organization of the event was great.
- Tools and accessibility were up-to-date and recommendations from the organizers in general were helpful.
- It would be great if both of our mentors where on site, but apart from that the communication, insights and feedback were very helpful!

Was it worth it?

- **Was this worth it?**

The opportunity to have access and work on such an impressive machine (B200), being able to learn from experienced mentors and also interacting with all the other participants is an experience that we wont encouter anywhere else and differently worth it. The industry lens was really eye-opening for us

- **Will you continue development?**

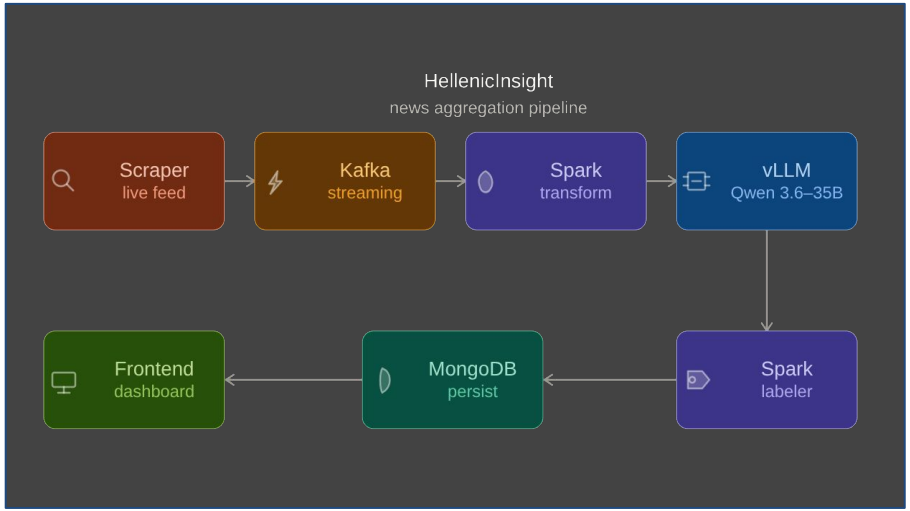
Yes, we are strongly considering it. Our application is already very close to being production-ready, as we use a contemporary tech stack. With the implementation of a few additional features, it can seamlessly transition into a fully complete, real-world platform.

- **What sustained resources/support will be critical for your work after the event?**

To bring this to full production, our critical needs include continued access to high-performance GPU cloud compute (for running our LLMs efficiently), ongoing mentorship regarding go-to-market and scaling strategies, and potentially seed funding to cover initial server hosting and database costs.

Application Background

- Real-time Greek News Aggregator Pipeline, detecting political bias of articles using fine-tuned LLM, first of its kind in Greece
- Important for raising awareness in the political field, due to the political influence towards media
- Heavy computational intensity for training the model, light on deployment and inference with vLLM



Hackathon Objectives and Approach

- Benchmarked a number of models in GreekMMLU
- Fine-Tuned Qwen 3.6 35B
- Real-Time streaming pipeline with Apache Kafka and Spark
- Hugging Face, Unsloth, vLLM

Technical Accomplishments and Impact

- **What:** Distilled frontier-model reasoning into a fine-tuned 35B model, achieving **~10% F1 boost** on Greek political bias classification
- **How:** LoRA on Qwen 3.6, labeled via Foundation Model Knowledge Distillation, served through **Scrapy** → **Kafka** → **PySpark** → **vLLM**
- **Why it matters:** Fully local, no API costs or external dependencies — architecture designed for high-throughput concurrent inference on HPC hardware

Please use around 100 words to summarize your team's achievements during this Hackathon.

During this Hackathon, our team engineered Hellenic Insight, a full-stack AI platform that quantifies political bias in Greek digital media. Demonstrating technological superiority, we utilized Crawl4AI for asynchronous data extraction, vLLM hosting Qwen for high-throughput inference, and a Kafka-to-PySpark-to-MongoDB pipeline. We showcased immense progress, evolving from basic scripts to a fully routed, bilingual React dashboard. Our solution boasts a high innovation quotient by mapping deep-structure ideological framing rather than basic sentiment. Utilizing open-source models, our architecture guarantees high viability, offering a cost-effective, highly scalable tool ready for real-world deployment that could be used by everyone.

Demo

Our Platform!

Generally Centered news article

Generally Politically Biased news article

PROMOTING YOUR WORK: AVAILABLE OPPORTUNITIES

- **Papers and Talks:** Please acknowledge the Open Hackathons program and OpenACC Organization in any planned or upcoming papers, presentations, or talks.
  "This work was completed in part at the [Event name], part of the Open Hackathons program. The authors would like to acknowledge OpenACC-Standard.org for their support."
- **Social Media Support:** Please feel free to promote your participation across your social media channels. Tag **@OpenACCorg** and **#OpenHackathons** and we are happy to amplify.
- **Blogs and Technical Write-ups**: Create a blog post or technical article that highlights the work being done and results achieved.
- **Quotes and Testimonials:** Highlight your quote or feedback on our channels (i.e. social, website, etc.).

***Please reach out to Izumi Barker (ibarker@nvidia.com) to discuss marketing options and opportunities.**